# Paul Parisot

Software Engineer - Core C/C++ & Application Security

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## About

Low-level systems engineer (C/C++, Assembly) specializing in performance-critical applications and memory safety.
Strong background in detecting and mitigating software vulnerabilities (buffer overflows, race conditions, memory leaks).
Experienced in Agile/TDD environments, with a practical understanding of relational databases and scripting for automation.

## Skills

**Core Languages**: C, C++, Assembly, Python, Shell Unix, SQL
**Data & Systems**: Relational Databases (Query optimization, Joins), Linux Environments
**Security & Testing**: TDD, Vulnerability Remediation, Pentesting (HackTheBox, Nmap, Privilege Escalation)
**Workflow**: Agile/Scrum, Jira, Git, CI/CD, DevOps (Docker)

## Professional Experience

**Gno.land - Core Software Engineer (Open-Source System)**

- Diagnosed and patched **critical memory-management vulnerabilities** (preventing panic/segfault scenarios).
- Enforced strict memory safety protocols and optimized concurrent processes to eliminate race conditions.
- Integrated extensive test suites under strict TDD and CI/CD policies, contributing **10k lines of code**.

**Faurecia / Forvia - Software Engineer (Embedded C++ / Agile)**

- Developed robust C++ network modules for embedded communication, ensuring secure low-level data exchange.
- Engineered a real-time navigation system showcased at **CES 2024**, delivering production-ready features ahead of schedule.
- Operated within an **Agile environment**, managing tasks via **Jira** and implementing rigorous code quality standards.

**Epitech / IONIS - Technical Instructor (Systems & C/C++)**

- Taught system-level programming and low-level architecture (C, C++, Asm) to over **400 students**.
- Evaluated code quality, focusing on memory optimization, structural precision, and the avoidance of runtime vulnerabilities.
- Produced clear technical documentation and structured assessment materials for complex programming concepts.

## Technical Projects & Security

**Sledge – Cloud Provider Migration System**

Developed a generic migration tool involving complex data structure handling and database interactions.
Applied strong relational algebra principles to optimize **SQL queries** and manage cross-platform data integrity.

**Skills:** SQL Optimization, Relational Databases, Data Architecture, Scripting

**Applied Security & Vulnerability Research**

Participant in CTF challenges (HackTheBox, TryHackMe) focusing on boot2root and privilege escalation.
Analyzed binary decompilation and low-level software defects, gaining a deep comprehension of system architectures, Assembly, and memory layouts.
Practical experience with vulnerability detection tools (Nmap, Gobuster, Metasploit, ...) and network security (Port forwarding, Netcat, ...).

**Skills:** Pentesting, Reverse Engineering, Threat Analysis, Network Protocols

**rodox – Modular Systems Engine**

Designed a low-level engine architecture in C/C++ natively for **Linux**.
Implemented highly optimized custom memory allocators to prevent fragmentation and ensure runtime stability.

**Skills:** C++, Linux, Custom Memory Management, Pointer Arithmetic, Low-level Optimization

## Education

**Epitech, Paris** 2022 – 2027
Master's Degree in Information Technology (Level-7 RNCP Certification)
Specialization: Systems Engineering and Software Architecture

**McGill University, Montréal** 2025 – 2026
Certificate in Management
Core coursework: Business Administration, Organizational Management